

FOUNDATION OF COMPUTING THEORY - HOMEWORK 01

Q.1) $S \rightarrow aA \epsilon \mid \epsilon Bb$

$C \rightarrow aCb \mid \epsilon$

$A \rightarrow a \mid aA \mid \epsilon$

$B \rightarrow b \mid bB \mid \epsilon$

Q.2) $S \rightarrow AB$

$A \rightarrow aAb \mid \epsilon$

$B \rightarrow \cancel{bBc} \mid bBc \mid \epsilon$

Q.

Q.3) $S \rightarrow aSa \mid bSb \mid a \mid b \mid \epsilon$

Q.4) $S \rightarrow aSa \mid bSb \mid A \mid B$

$A \rightarrow aAb \mid bAa \mid aBb \mid bBa$

$B \rightarrow aBa \mid bBb \mid a \mid b \mid \epsilon$

Q.6) Given grammar: $S \rightarrow aSb \mid aSb \mid \epsilon$

a) language, $G = (V, \Sigma, R, S)$

where,

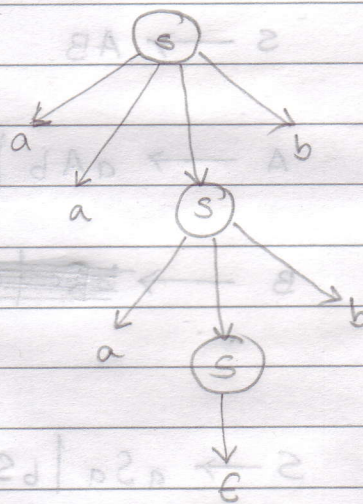
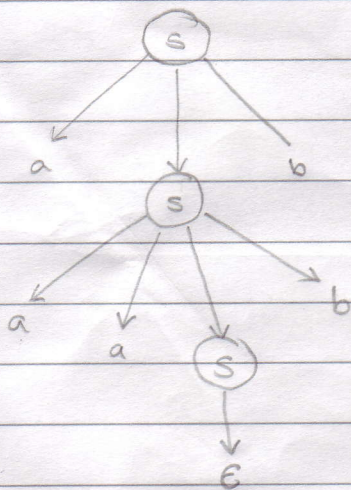
$$V = \{S\}$$

$$\Sigma = \{a, b\}$$

$$R = \{S \rightarrow aSb, S \rightarrow aasb, S \rightarrow \epsilon\}$$

$$S \in V.$$

b) There are two possible parse trees for the string 'aaabbb'. Hence the CFG is ambiguous.



a) $L(G) = \{a^i b^j \mid j \leq i \leq 2j, i, j \geq 0\}$

c) $S \rightarrow aSb \mid A$
 $A \rightarrow aaAb \mid \epsilon$

Q.5) - let us assume that CFG's are closed under complement

→ Then $A \cap B$

- We know that CFGs are ^{not} closed under intersection

- Hence

$$A \cap B = \overline{\overline{A} \cup \overline{B}}$$

where A and B are languages that are CFGs.

$$A = \text{CFG}$$

$$B = \text{CFG}$$

$$\overline{A} = \text{CFG} \dots \text{from assumption}$$

$$\overline{B} = \text{CFG} \dots \text{from assumption}$$

$$\overline{A} \cup \overline{B} = \text{CFG} \dots \text{CFGs are closed under union}$$

$$\overline{\overline{A} \cup \overline{B}} = \text{CFG} \dots \text{from assumption}$$

$$\therefore A \cap B = \text{CFG}$$

But this is not true and hence our assumption is ~~wrong~~

Hence, CFGs are not closed under complement.